Advanced HCI

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Input Modalities

Seeing – Image (CNNs)

- Enables visual interaction through gestures, facial expressions, and eye-gaze tracking.

Hearing & Talking – Sound

- Supports speech recognition, emotion detection, and environmental audio interaction.

Reading & Writing – Text (NLP)

- Allows natural language interaction via chatbots, virtual assistants, and text-based systems.

Interactive Technologies & Interfaces

Adaptive Interfaces

Adjust dynamically to user abilities, preferences, and context.

Persuasive Interfaces

Designed to ethically influence user behavior through feedback, motivation, and engagement strategies.

Inductive Interfaces

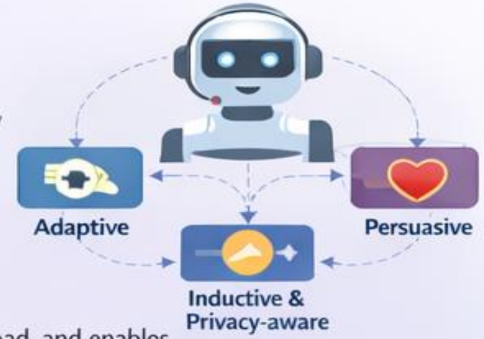
Encourage users to learn through exploration and discovery, enhancing learnability.

Privacy-aware Interfaces

Communicate and give control over sensitive user data, integrating ethics into interface design.

Agentic AI

- **Definition:** Autonomous and proactive AI systems that act on behalf of users while keeping them in control.
- **Support for Interfaces:**
  - **Adaptive Interfaces:**
    - Observes user behavior, and context to automatically personalize content, layout, or interaction strategies.
  - **Persuasive Interfaces:**
    - Monitors progress and delivers timely, targeted nudges or suggestions ethically.



- **HCI Impact:** Enhances usability, reduces cognitive load, and enables proactive, intelligent interaction while maintaining transparency and trust.

Evaluation & User Testing

Usability Testing

Empirical assessment of efficiency, effectiveness, and satisfaction of systems with real users.

Task Analysis & KLM

Methods to understand user behavior, model tasks, and design efficient, predictable interactions.

Evaluation

Applies qualitative and quantitative methods to measure performance, experience, accessibility, and impact.

Establishing the Sensory Layer

1. Input Modalities

Seeing – Image (CNNs)

Enables visual interaction through gestures, facial expressions, and eye-gaze tracking.

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Supports speech recognition, emotion detection, and environmental audio interaction.

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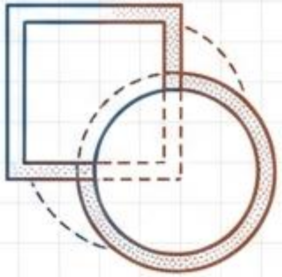
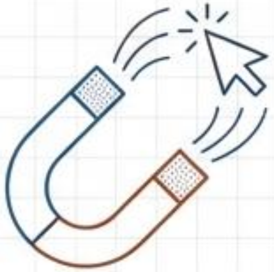
Interaction Frameworks and Taxonomies

2. Interactive Technologies and Interfaces



Adaptive Interfaces

Adjust dynamically to user abilities, preferences, and context to improve usability and satisfaction.



Persuasive Interfaces

Designed to ethically influence user behavior through feedback, motivation, and engagement strategies.

Inductive Interfaces

Encourage users to learn through exploration and discovery, enhancing learnability.

Privacy-aware Interfaces

Communicate and give control over sensitive user data, integrating ethics into interface design.

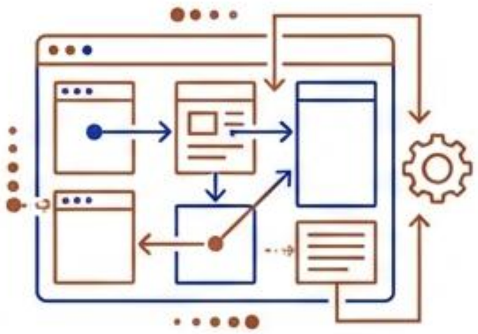
The Core Logic of Agentic Systems
3. Agentic AI

Definition: Autonomous and proactive AI systems that act on behalf of users while keeping them in control.

Support for Interfaces

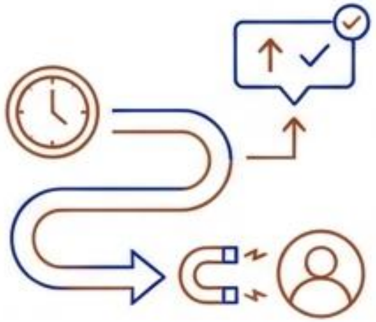
Adaptive Interfaces

Observes user behavior and context to automatically personalize content, layout, or interaction strategies.



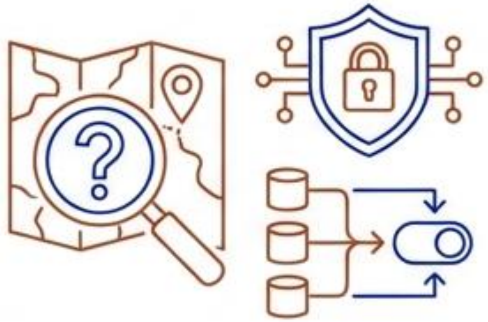
Persuasive Interfaces

Monitors progress and delivers timely, targeted nudges or suggestions to guide user behavior ethically.



Inductive & Privacy-aware Interfaces

Can provide contextual hints, assist discovery, or ensure data usage aligns with user expectations.



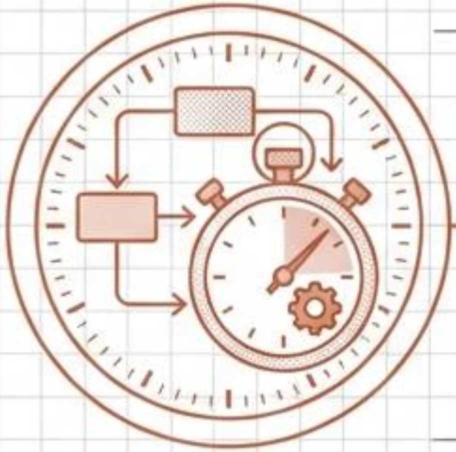
HCI Impact: Enhances usability, reduces cognitive load, and enables proactive, intelligent interaction while maintaining transparency and trust.

Verification and Performance Metrics

4. Evaluation and User Testing

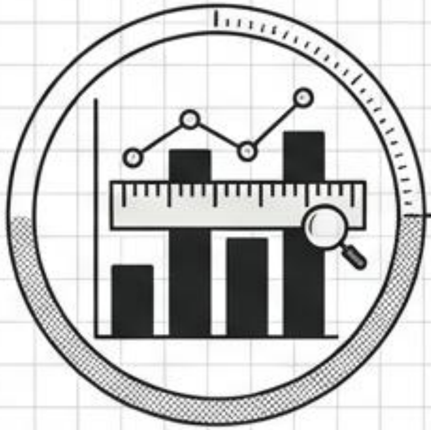
Usability Testing

Empirical assessment of efficiency, effectiveness, and satisfaction of systems with real users.



Task Analysis & KLM

Methods to understand user behavior, model tasks, and design efficient, predictable interactions.



Evaluation

Applies qualitative and quantitative methods to measure performance, experience, accessibility, and impact, ensuring evidence-based design.

Advanced HCI Lab Topics

- Python basics for HCI
- Image Processing with CNNs
- Sound Processing
- NLP Tasks
  - Agentic AI:
    - Perception Agent
    - Questions Agent
    - Adaptive Agent
    - Hints and Explanation Agent
  - Orchestrate / Manager Agent

- Perception Agent
- Questions Agent
- Adaptive Agent
- Hints and Explanation Agent
- Orchestrate / Manager Agent
- Data Analysis with Python

Lab Topics / Activities

- **Week 1&2: Python basics for HCI: Variables, functions, and libraries** (Note: Information regarding specific Python introductory curriculum is external to the provided sources).

- **Week 3: Implement a simple image processing tasks using CNNs** This lab focuses on the "Seeing" modality, where **Convolutional Neural Networks (CNNs)** enable visual interaction through gestures, facial expressions, and eye-gaze tracking.

- **Week 4: Implement a simple sound processing: audio feature extraction and speech recognition** Students will work with the "Hearing & Talking" modality, which supports **speech recognition**, environmental audio interaction, and emotion detection.

- **Week 5: Implement a simple NLP tasks** This session utilizes **Natural Language Processing (NLP)** to allow for natural language interaction via text-based systems.

Lab Topics / Activities

- **Week 6-9: Agentic AI: create a simple adaptive agentic AI**

- **Week 10 and 11: Data Analysis with Python – Implement basic statistics and analyse collected interaction data to evaluate system performance** This final lab applies **Evaluation** methods, using quantitative and qualitative data to measure performance, experience, and accessibility to ensure evidence-based design.